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11.4.4 Mathematical functions (math.h)

Most of the mathematical functions are defined in math.h header file which includes basic mathematical functions.

11.4.4.1 cos() function

The cos() function takes a single argument in radians. The cos() function returns the value in the range of [-1, 1]. The returned value is either in double, float, or long double.

Program 11.12

```
#include <iostream>
#include <math.h>
using namespace std;
int main()
{
        double x = 0.5, result;
        result = cos(x);
        cout << "COS("<<x<<")= "<<result;
}
```

Output:

```
COS(0.5)= 0.877583
```

11.4.4.2 sqrt() function

The **sqrt()** function returns the square root of the given value of the argument. The **sqrt()** function takes a single non-negative argument. If a **negative value** is passed as an argument to **sqrt()** function, a **domain error occurs**.

Program 11.13

```
#include <iostream>
#include <math.h>
using namespace std;
int main()
{
        double x = 625, result;
        result = sqrt(x);
        cout << "sqrt("<<x<<") = "<<result;
        return 0;
}
```

Output:

```
sqrt(625) = 25
```